Notes: Wang and Yang (JPE, 2025)

Policy Experimentation in China: The Political Economy of Policy Learning

Contents

1 Big picture

- **Question.** How does large-scale policy experimentation actually work in China, and does it generate unbiased policy learning for the central government?
- **Setting.** The paper studies China's centrally coordinated policy experimentation system from 1980 to 2020. The core institution is the use of *experimentation points* (*Shidian*), where the central government pilots a policy in selected localities before deciding whether to roll it out nationally.
- **Core challenge.** Experimentation is supposed to teach the central government about whether a policy works on average. But experiments may be run in places that are not representative, under incentives that are not representative, and interpreted by decision-makers who do not fully filter out confounding shocks.
- **Main descriptive result.** The authors build a comprehensive database of 652 policy experiments using 19,812 central and local government documents. Of these experiments, 42.0% eventually became national policies.
- **Main substantive result: three facts.**
  - **Positive sample selection.** Experimentation sites are not representative. In 87.7% of experiments, sites are positively selected on local economic conditions; experimentation sites are on average 44.2% richer in fiscal revenue than nonexperimentation sites.
  - **Nonrepresentative experimental situations.** Local politicians exert extra effort during experiments because successful experiments help promotion. Domain-specific fiscal expenditure rises by about 1.3% during experimentation, especially when politicians have stronger career incentives, but this extra effort does not persist when the policy rolls out nationally.
  - **Naive interpretation of outcomes.** The central government is not fully sophisticated in evaluating experiment results. Exogenous land-revenue windfalls and exogenous changes in local politicians' career incentives during the experiment increase the probability that a policy is rolled out nationally, even though these shocks are orthogonal to the policy's intrinsic effectiveness.
- **Main implication.** If the government's objective is to learn the policy's average treatment effect (ATE), then these three facts imply biased policy learning. The paper shows that naive experimentation outcomes predict national rollout, but more careful estimates that account for pretrends or use synthetic control do not.
- **Main scaling result.** Among economic policies that are eventually rolled out nationally, 71.1% exhibit smaller effects during national rollout than during experimentation. This is the paper's empirical signature of shrinkage from unrepresentative experiments.
- **Why the paper matters.** It turns China's famous gradualism into a measurable political-economy object. The paper shows that experimentation can help reform, but it can also generate distorted evidence when site selection, political incentives, and interpretation are not well designed.

2 Data and measurement (key objects)

2.1 Government-document panel on policy experimentation

- **Main source.** The core dataset is built from PKU Law, a comprehensive archive of Chinese government documents.
- **Search rule.** The authors collect the near-universe of central and local government documents from 1980 to 2020 containing the keywords *experimentation points* (*Shidian*) and *experimental zones* (*Shiyanqu*).
- **Raw document count.** The data contain 19,812 documents in total:
  - 4,399 central-government documents,
  - 15,413 local-government documents.
- **Experiment definition.** These documents are grouped into 652 distinct policy experiments based on policy theme. Closely related simultaneous experiments can be combined into one broader experiment.
- **Who initiates experiments?** The 652 experiments are initiated by 92 central ministries and commissions; 98 ministries and commissions are involved overall once collaborators are counted.
- **Intended scope.** Of the 652 experiments:
  - 613 are national experiments intended for potential national rollout,
  - 39 have explicit regional targets.
- **Rollout status.** Overall, 42.0% of experiments eventually become national policies and 58.0% do not.

2.2 Experiment-level characteristics

- **Rounds and sites.** Experiments often unfold in multiple waves. The paper identifies 1,374 rollout rounds across the 652 experiments.
- **Average scale.** The average experiment has:
  - 2.1 rounds,
  - 19.0 experimentation sites,
  - duration of about 2.25 years.
- **Participation rule.** About 42.6% of experiments allow at least partial voluntary participation by local governments; the rest are directly assigned by the center.
- **Administrative level.** Experiments can be implemented at province, prefecture, or county level. This matters because representativeness is tested at the relevant administrative level.
- **By level.** Among national experiments:
  - Province-level experiments: 199 experiments, 1.7 rounds, 4.9 sites, 36.18% rollout.
  - City/county-level experiments: 414 experiments, 2.3 rounds, 26.8 sites, 47.34% rollout.

2.3 Policy domains, uncertainty, and complexity

- **Policy domains.** Experiments are classified into broad domains such as resource/environment, market supervision, agriculture, education, finance, tax/fiscal policy, health, commerce, industry, development and reform, labor, and transportation.
- **Ex ante uncertainty.** The paper codes several proxies for whether the policy is already viewed as likely to become national:
  - whether the center lays out a detailed national rollout timeline before experimentation,
  - whether implementation details are drafted in advance,
  - whether the policy appears in a Five-Year Plan,
  - the degree of academic consensus around the policy before experimentation.
- **Ex ante complexity.** The paper codes several proxies for complexity:
  - whether multiple ministries are involved,
  - whether central documentation is long or split across multiple documents,
  - whether experimentation lasts a long time,
  - whether many local implementation documents are issued.
- **Examples.**
  - Carbon emission trading: initiated in 2011, 7 sites, rolled out nationally in 2021.
  - Separation of permits and licenses: initiated in 2015, 24 prefectures in 3 waves, rolled out nationally in 2018.
  - Agricultural catastrophe insurance: initiated in 2017, 14 provinces over 2 waves, not rolled out nationally.
  - County fiscal empowerment reform: 2002–2015, 1,246 counties over more than 10 waves, not rolled out nationally.

2.4 Auxiliary locality- and politician-level data

- **Local socioeconomic conditions.** The main locality variables include local GDP, GDP per capita, fiscal revenue, fiscal expenditure, and domain-specific public spending.
- **Politician characteristics.** The authors collect biographical information on ministers and on provincial/prefectural leaders over four decades.
- **Career incentives.** A politician's career incentive is proxied by an ex ante promotion probability projected from starting age of tenure, bureaucratic rank, and political position.
- **Other local shocks.** The paper also measures land-conveyance revenue, land suitable for construction, national interest rates, and local unrest events.

2.5 Unobservables and timing

- **Timing.** The analysis separates:

- preexperiment characteristics of sites,
- politician effort and local fiscal allocation during experimentation,
- national rollout decisions after experimentation,
- national outcomes after rollout.

- **Unobservables.** The main latent objects are:
  - the true policy effect in a representative locality,
  - unobserved local political effort during the experiment,
  - shocks to local outcomes unrelated to the policy itself.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework: observed experimentation outcomes

The paper defines observed experimentation outcomes as

$$\widehat{Y}(p, I_t, E_t),$$

where p is the policy on trial, I_t is the vector of preexperimentation local characteristics of selected sites, and E_t is the vector of local politicians' incentives and efforts during the experiment.

The observed outcome is decomposed as

$$\widehat{Y}(p, I_t, E_t) = Y(p, \bar{i}_t, \bar{e}_t) + F_{i,p}(I_t - \bar{i}_t) + F_{e,p}(E_t - \bar{e}_t) + G_{i,e}(I_t, E_t). \tag{1}$$

- $Y(p, \bar{i}_t, \bar{e}_t)$ is the **ATE-like object**: the average effect of policy p when implemented in a representative locality with representative politician incentives and effort.
- $F_{i,p}(I_t - \bar{i}_t)$ is the **site-selection component**. If experimentation sites are richer or better governed than average, outcomes during experimentation can exceed average rollout outcomes even if the policy itself is unchanged.
- $F_{e,p}(E_t - \bar{e}_t)$ is the **experimental-situation component**. If politicians work unusually hard during experiments because of promotion incentives, observed experiment effects overstate what the country will get after rollout.
- $G_{i,e}(I_t, E_t)$ collects **other shocks** that improve measured outcomes during experimentation but are not policy effects. The leading examples in the paper are local land-revenue windfalls and politician turnovers that change local leaders' career incentives.

Interpretation. Equation (1) is the core logic of the paper. It says that the experiment outcome the center observes is not the same thing as the policy's representative effect unless sites are representative, incentives are representative, and unrelated shocks are stripped out.

3.2 Conceptual framework: central government's rollout decision

The paper writes the center's rollout decision as

$$D\Big(\widehat{Y}(p, I_t, E_t) - d_1[F_{i,p}(I_t - \bar{i}_t) + F_{e,p}(E_t - \bar{e}_t)] - d_2\, G_{i,e}(I_t, E_t) + d_3 Z_{pit}\Big)\,, \tag{2}$$

where Z_{pit} represents other objectives the center may care about, such as political stability.

- If the center wants to learn the policy's average effect, it should set $d_1 = 1$: fully net out site selection and extra effort.
- If the center is fully sophisticated about unrelated shocks, it should also set $d_2 = 1$.
- If $d_3 \neq 0$, the center is optimizing over more than average economic outcomes, for example, unrest avoidance or other political objectives.

Interpretation. The empirical sections ask whether actual rollout choices look like they come from a decision rule with $d_1 = 1$ and $d_2 = 1$. The answer is basically no.

3.3 Testing representativeness of experimentation sites

For each policy experiment i, the paper compares preexperiment characteristics between experimentation and nonexperimentation sites and computes a Student t-statistic:

$$t_i = \frac{\widehat{Y}_i(1) - \widehat{Y}_i(0)}{\sqrt{\widehat{S}_i^2(1)/n_{i,1} + \widehat{S}_i^2(0)/n_{i,0}}}. \tag{3}$$

- $\widehat{Y}_i(1)$ is the average characteristic of experimentation sites.
- $\widehat{Y}_i(0)$ is the average characteristic of comparable nonexperimentation sites.
- Positive t_i means experimentation sites are better endowed than the comparison pool.

Interpretation. This is the paper's reduced-form measure of selection bias. The main baseline characteristic is local fiscal expenditure or revenue in the year before experimentation begins.

3.4 Main empirical result on site selection

The representativeness exercise yields three central facts.

- The distribution of t-statistics is overwhelmingly positive.
- The average t-statistic is 5.01 in the full sample and 5.17 among national experiments.
- Only 43.0% of all experiments are classified as representative under the paper's conservative criterion, so most are not representative.

The paper also documents that province-level experiments are much closer to representative than city/county-level experiments. This is important because province-level sites are more directly selected by the center, whereas lower-level sites are often selected by provincial governments.

3.5 Promotion incentives and successful experiments

The paper first studies whether politicians are rewarded for successful experimentation. The exact reduced-form regression is not central; the main outcome is local leader promotion and the key regressor is the number of successful experiments a leader participated in during tenure.

- Participation in experiments per se does not predict promotion.
- Participation in *successful* experiments does.
- One successful experiment is associated with a 23.5% increase in promotion probability.
- This reward is larger in small-scale experiments, where fewer politicians share the credit.

Interpretation. Promotion incentives create a reason for politicians to overexert effort during experimentation.

3.6 Triple-differences specification for strategic fiscal effort

To test for extra effort during experimentation, the paper estimates

$$y_{ikt} = \alpha + \beta\, Exp_{ikt} + \lambda_{it} + \delta_{kt} + \nu_{ik} + \varepsilon_{ikt},$$

using county-domain-year data.

- y_{ikt} is the share of county i's fiscal expenditure going to domain k in year t.
- Exp_{ikt} is the number of experiments in domain k that county i participates in during year t.
- λ_{it}: county-by-year fixed effects.
- δ_{kt}: domain-by-year fixed effects.
- ν_{ik}: county-by-domain fixed effects.

Interpretation. This triple-differences setup asks: when a county starts an experiment in a particular policy domain, does it raise spending *in that domain*, relative to the county's overall spending trend and the national trend in that domain?

3.7 Main result on fiscal effort

The coefficient β is positive and economically meaningful:

- An additional experiment increases the expenditure share in the corresponding fiscal domain by about 1.3%.
- The increase is stronger for local politicians with higher career incentives.
- There is no pretrend before experimentation begins.
- The increase is absent for nonexperimentation sites when the same policy later rolls out nationally.
- Within experimentation sites, the extra spending stops when experimentation ends.

Interpretation. The experiment creates extra effort that is targeted to the pilot itself and does not survive scaling.

3.8 Additional measure of strategic effort: differentiated implementation

The paper also measures whether local governments try to stand out by making their implementation documents more distinct from those of other participating sites.

- It constructs pairwise text-similarity measures using latent semantic analysis.
- Politicians with stronger career incentives issue more differentiated experimentation documents.

Interpretation. This is another form of strategic effort: not just spending more, but trying to become a model site through differentiated implementation.

3.9 Locality-specific shocks and the rollout decision

To test whether the center filters out unrelated local shocks, the paper uses exogenous land-revenue windfalls during experimentation. The two-stage least squares system is

$$LandRevenue_{ipt} = \alpha\, Suitability_i \times Interest_t + X_{it}'\beta + d_i + g_t + d_m + e_{ipt},$$

$$y_p = \mu\, \widehat{LandRevenue}_{ipt} + X_{it}'\Gamma + \omega_i + \eta_t + d_m + \varepsilon_{ipmt}.$$

- $LandRevenue_{ipt}$ is local land-conveyance revenue at experimentation site i for policy p in year t.
- $Suitability_i \times Interest_t$ is the instrument:
  - $Suitability_i$ measures terrain suitable for construction (slope $\leq 15°$),
  - $Interest_t$ is the national interest rate, which shifts macro housing demand.
- y_p is an indicator that policy p eventually rolls out nationally.

Interpretation. If land-revenue windfalls during experimentation make rollout more likely, then the center is reading local macro luck as if it were policy success.

3.10 Main IV result on naive interpretation of local shocks

The first stage is very strong, with an F-statistic around 623. The second stage shows that higher instrumented land revenue during experimentation raises the probability of national rollout.

- This effect survives county, year, and ministry fixed effects.
- Placebo slopes above the legal construction cutoff do not predict land revenue.
- Land-revenue shocks *after* experimentation ends have no effect on rollout.

Interpretation. The center appears to mistake exogenous local revenue windfalls for successful policy outcomes.

3.11 Politician-specific shocks and rollout

The second class of naive interpretation uses politician turnover during the experimentation period:

$$y_p = \alpha\, Turnover_{ip} + \beta_1\, Turnover_{ip} \times IncreaseIncentive_{ip} + \beta_2\, Turnover_{ip} \times DecreaseIncentive_{ip} + g_t + d_m + v_n + \varepsilon_{ipm}$$

- $Turnover_{ip}$ indicates a change in the party secretary in experimentation locality i during policy p's experimentation period.
- $IncreaseIncentive_{ip}$ and $DecreaseIncentive_{ip}$ measure whether the incoming leader has higher or lower promotion incentives than the outgoing one.
- y_p is again the indicator that policy p rolls out nationally.

Interpretation. If exogenous politician turnover that changes career incentives affects rollout, then the center is again partly rewarding politician-specific shocks rather than policy quality.

3.12 Main turnover result

The paper finds:

- Turnover itself has no significant effect.
- If the incoming politician has stronger upward mobility than the outgoing politician, national rollout becomes substantially more likely.
- If turnover lowers politicians' career incentives, rollout becomes less likely.
- The effect is stronger in small-scale experiments and robust to alternative measures of political incentives.

Interpretation. The center does not fully strip out politician-specific shocks that alter experimentation outcomes.

3.13 Naive versus sophisticated experimentation-effect estimators

The paper compares three ways of estimating experimentation effects:

- **Naive pre/post estimator:** average GDP per capita after the experiment minus before the experiment among experimentation sites.
- **Pretrend-adjusted estimator:** controls for province-specific trends.
- **Generalized synthetic control estimator:** matches each experimentation site to a weighted combination of nonexperimentation counties with similar pretrends.

Key result.

- The naive pre/post estimator strongly predicts national rollout.
- The pretrend-adjusted and synthetic-control estimators do not.

More concretely, a one-standard-deviation increase in the naive experimentation effect is associated with a 7.0 percentage-point (17.9%) higher probability of national rollout.

Interpretation. The center behaves as if it uses raw observed experiment outcomes, not carefully debiased experiment outcomes.

3.14 Rollout-stage heterogeneity by similarity to experimentation sites

To study national policy outcomes after rollout, the paper estimates

$$Growth_{cpt} = \alpha\, M_{cp} + g_c + j_t + h_p + \varepsilon_{cpt},$$

where $Growth_{cpt}$ is GDP per capita growth in nonexperiment county c after policy p rolls out nationally, and M_{cp} is the Mahalanobis distance between county c and the original experimentation sites.

- In panel A, M_{cp} is computed from socioeconomic characteristics.
- In panel B, M_{cp} is computed from politicians' career incentives.

Interpretation. If policies learned from unrepresentative experiments scale badly, then counties more similar to experimentation sites should benefit more after national rollout.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

The main failure is to treat experimentation outcomes as if they were direct measures of a policy's average effect. That interpretation ignores three problems:

- experimentation sites may be positively selected,
- local politicians may exert extra effort during pilots,
- unrelated local or politician-specific shocks may contaminate observed outcomes.

The whole paper is designed to separate these three channels.

4.2 Why preexperiment comparisons identify site selection

The selection test is clean because it compares *predetermined* local characteristics before the experiment begins.

- Reverse causality is not a concern because the comparison uses preexperiment data.
- The main remaining concern is whether relevant comparison units are chosen correctly. The paper addresses this by stratifying at the correct administrative level and by running many alternative specifications.
- The positive-selection pattern survives alternative local characteristics, policy-domain matching, weighting rules, and permutation-style robustness checks.

4.3 Why province versus city/county comparisons are informative

Province-level experiments are more directly selected by the central government, while prefectural and county experiments are often selected by provincial governments conditional on the province being chosen.

- Province-level experiments are much less positively selected.
- Lower-level experiments are much more positively selected.

Why this matters. This contrast is strong evidence that misaligned objectives between central and local governments are part of the selection problem.

4.4 Why the triple-differences specification identifies strategic effort

The triple-differences regression identifies whether counties reallocate spending specifically toward the domain of the policy being piloted.

- County-by-year fixed effects absorb the county's overall spending conditions in that year.
- Domain-by-year fixed effects absorb national changes in that domain.
- County-by-domain fixed effects absorb time-invariant county-domain propensities.

So the remaining variation is exactly the domain-specific increase in spending caused by policy experimentation in a particular county and year.

4.5 Why the land-suitability instrument is credible

The instrument interacts a predetermined geographic constraint with a national macro shock.

- Land suitability is fixed by terrain.
- National interest rates are determined at the macro level and are plausibly exogenous to an individual county.
- Only land below the legal slope threshold should matter for construction; placebo slopes do not work.
- Land revenue realized after experimentation ends does not affect rollout.

These facts make the IV strategy persuasive as a source of exogenous local fiscal windfalls.

4.6 Why politician turnover is informative

The paper focuses on local-leader turnover after the experiment begins.

- This timing reduces concern that the local government's original participation decision is endogenous to later career incentives.

- The Organization Department, not the policy ministry, determines political rotation, so the turnover is unlikely to be tailored to a specific policy experiment.
- The effects are strongest when turnover changes career incentives and absent when turnover occurs before experimentation or long after it begins.

Interpretation. The turnover evidence gets as close as the paper can to exogenous variation in politician effort during experimentation.

4.7 Why the rollout-prediction exercise is informative

The paper does not observe the center's objective function directly. Instead, it infers it from which experimentation-effect estimator best predicts national rollout.

- If the center cared about average policy effects, corrected estimators should predict rollout better.
- In fact, only the naive estimator predicts rollout.

This is clever because it converts the hidden policymaker decision rule into an observable empirical question.

4.8 Relation to the literature

- **Policy experimentation and reform.** The paper speaks directly to the China gradualism literature, especially work emphasizing *experimentation under hierarchy*.
- **External validity and scaling.** It connects to the scaling literature by showing both *sample-selection bias* and *nonrepresentative implementation effort* in a real-world government experimentation system.
- **Political incentives.** It complements the literature on career concerns and political tournaments in China by showing a downside: stronger promotion incentives improve experimentation outcomes but also bias policy learning.
- **State capacity and policy design.** It shows that centralized experimentation can avoid some problems of decentralized underexperimentation, but introduces its own informational distortions.

5 Tables and figures

5.1 Figure 1 and Table 1: scale and evolution of policy experimentation

Read:

- Figure 1 shows a hump-shaped time pattern in new policy experiments. The center launches relatively few in the 1980s and 1990s, the number rises sharply in the late 1990s and 2000s, peaks at 47 new experiments in 2010, and then declines.
- The success rate remains broadly stable over time: roughly 42% of experiments become national policies and 58% do not.

- Table 1 shows that the average experiment has 2.1 rounds and 19.0 sites. National experiments are larger than subnational ones.
- Province-level experiments are much more representative than city/county-level ones, which is one of the paper's first clues that lower-level site selection is strongly distorted.

5.2 Figure 2: where experimentation happens and concrete examples

Read:

- Figure 2A maps the total number of experiments each province participated in between 1980 and 2020.
- The map is visibly uneven: experimentation is concentrated in richer, coastal, and administratively important areas.
- Panels B.1–B.4 are useful because they show that some policies scale quickly from a handful of rich pilot sites to the entire country, while others expand in waves and still never become national.
- The county fiscal empowerment reform is especially revealing: it involved 1,246 counties and more than 10 waves, yet still never became a national policy.

5.3 Figure 3: experimentation sites are positively selected

Read:

- Figure 3A plots the density of the representativeness t-statistics. The mass is overwhelmingly on the positive side.
- Figure 3B shows that the average t-statistic remains positive across a wide range of alternative specification choices.
- This is strong visual evidence that site selection is not a small nuisance issue. It is a first-order empirical fact.
- The conservative representativeness tests still imply that most experiments are not representative.

5.4 Table 2: local fiscal expenditure during policy experimentation

Read:

- Panel A shows that experimentation induces more spending in the policy's own fiscal domain. The key magnitude is about a 1.3% increase in the domain's share of total fiscal expenditure.
- The interaction with career incentives is positive: politicians with stronger promotion incentives increase fiscal support more aggressively.
- Panel B shows no analogous expenditure surge among nonexperimentation sites during national rollout.

TABLE 1
Summary Statistics of Policy Experimentation

	Number of Experiments (1)	Number of Rounds (2)	Number of Sites (3)	% Rollout (4)	Average t-Statistics (5)	% Representative (6)
			A. Full Sample			
Overall	652	2.1	19.0	42.02	5.01	43.0
National	613	2.1	19.7	43.72	5.17	41.9
National × completed	509	2.1	18.8	50.88	5.48	39.4
National × ongoing	104	2.1	23.9	8.65	3.58	56.3
Subnational	39	2.0	8.7	15.38	2.22	60.0
Subnational × completed	35	2.0	9.2	17.14	2.16	59.3
Subnational × ongoing	5	2.0	11.4	0.00	2.72	50.0
			B. By Policy Domain			
Resource, energy, and environment	80	2.2	11.5	38.75	3.94	57.1
Market supervision	79	1.9	10.9	44.30	5.91	33.9
Agriculture	60	2.1	39.4	33.33	3.91	56.6
Education	56	2.3	39.2	46.43	5.43	28.3
Finance	53	1.8	6.2	47.17	8.50	40.6
Tax and fiscal policy	41	2.2	10.2	53.66	5.38	38.2
Population and health	38	2.3	21.2	47.37	4.57	36.1
Commerce and trade	36	2.1	17.0	41.67	6.34	23.1
Industry and information technology	35	1.8	25.2	37.14	6.87	24.0
Domestic affairs	31	2.3	15.7	29.03	4.12	36.0
Development and reform	29	2.0	23.0	37.93	4.25	60.0
Labor	22	2.5	9.9	45.45	4.61	55.6
Transportation	20	2.0	9.2	55.00	3.09	58.8
Others	33	1.9	34.2	66.67	5.27	40.0
			C. By Administrative Level			
Province level	199	1.7	4.9	36.18	1.44	72.4
City and county level	414	2.3	26.8	47.34	6.57	31.5

Note.—This table reports the summary statistics for our policy experimentation sample. In panel A, we present information on all 652 experiments and disaggregate them by national experiments (613) and subnational ones (39). In panels B and C, we focus only on those national experiments.

TABLE 2
Local Fiscal Expenditure during Policy Experimentation

	Share of Fiscal Expenditure on Experimentation-Related Domains					
	(1)	(2)	(3)	(4)	(5)	(6)
	A. Fiscal Input among Experimentation Sites					
Number of experiments	.003***	.002***	.002***	−.013***	−.002*	−.002
	(.001)	(.0004)	(.0005)	(.003)	(.001)	(.002)
# × career incentive				.036***	.009***	.011***
				(.006)	(.003)	(.003)
	B. Fiscal Input among Nonexperimentation Sites during National Policy Rollout					
Number of rolled-out policies	.001	.001	.001	.001	.001	.001
	(.001)	(.0004)	(.001)	(.003)	(.001)	(.002)
# × career incentive				−.001	−.0004	−.0003
				(.005)	(.002)	(.003)
Number of observations	150,977	150,977	150,977	142,128	142,116	142,116
Number of clusters	1,973	1,973	1,973	1,973	1,973	1,973
Mean of dependent variable	.173	.173	.173	.173	.173	.173
County by category fixed effects	No	Yes	Yes	No	Yes	Yes
Year by county fixed effects	Yes	No	Yes	Yes	No	Yes
Category by year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Note.—This table estimates the impact of a policy experiment on the fiscal expenditures of its experimentation sites. We characterize six general fiscal domains and match each policy experiment to its most closely related domain. In panel A, we investigate whether the experimentation units reallocated fiscal resources to the corresponding fiscal domain when a policy experiment is assigned. The average number of policy experiments within each prefecture-year-domain grid is 0.218, and the standard deviation is 0.541. Career incentives are measured as the ex ante probability of promotion projected by the start age of tenure and hierarchical level (mean = 0.481; SD = 0.075). In panel B, we investigate whether the previously nonexperimentation sites exhibited similar fiscal reallocation in the year that the policy rolled out nationally. Standard errors (shown in parentheses) are clustered at the county level.

* $p < .10$.

*** $p < .01$.

- This table is the cleanest empirical evidence that experimentation changes implementation effort itself, not just where policies are tested.

5.5 Table 3: naive interpretation of experimentation outcomes

Read:

- Panel A shows that instrumented land-revenue windfalls during experimentation make national rollout more likely.
- The first stage is extremely strong, which helps credibility.
- Panel B shows that politician turnovers that raise local leaders' career incentives also raise national rollout probabilities.
- Taken together, these results say that the center does not fully filter out shocks that are orthogonal to the policy itself.

TABLE 3
Naive Evaluation of Policy Experimentation

	National Rollout		
	(1)	(2)	(3)
	A. Land Revenue Windfall		
Land revenue (instrumented)	.008***	.006***	.009***
	(.002)	(.001)	(.001)
First-stage *F*-statistics	670.34	636.36	622.90
Number of observations	66,128	66,128	66,128
Number of clusters	1,644	1,644	1,644
Mean of DV	.612	.612	.612
Experiment year fixed effects	Yes	Yes	Yes
County fixed effects	No	Yes	Yes
Ministry fixed effects	No	No	Yes
	B. Politicians' Incentive Changes due to Political Rotation		
Rotation	.004	.016	.017
	(.019)	(.014)	(.014)
Positive rotation × ΔIncentive	.679***	.539***	.508***
	(.089)	(.090)	(.093)
Negative rotation × ΔIncentive	−.496***	−.459***	−.407***
	(.162)	(.132)	(.141)
Number of observations	3,899	3,899	3,899
Number of clusters	27	27	27
Mean of DV	.321	.321	.321
Ministry fixed effects	No	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Province fixed effects	No	No	Yes

Note.—In this table, we investigate whether external shocks to a policy experiment's sites and the local officials affect its likelihood of being rolled out as a national policy. Panel A reports the second stage of a 2SLS regression where we use the interaction term between area of land unsuitable for agricultural use and national interest rate to instrument for the land revenue (in logarithm) received by the local government. The average log land value is 5.27, with a standard deviation of 3.97. Standard errors (shown in parentheses) are clustered at the county level. We report the first-stage results in table A.14. Panel B is an analysis focusing on political rotations that happened *after* the selection of experimentation sites. At the experiment-by-prefecture level, we calculate the difference in career incentives between the leaving prefectural official and his immediate successor. "Rotation" is a dummy variable indicating political turnover during the experimentation, which is defined to be the period between the start of the first round of experimentation and 2 years after the last round. An average positive rotation is accompanied with an incentive increase of 0.079 (SD = 0.076). An average negative rotation is accompanied by an incentive drop of 0.055 (SD = 0.061). The standard errors are clustered at the province level.
*** $p < .01$.

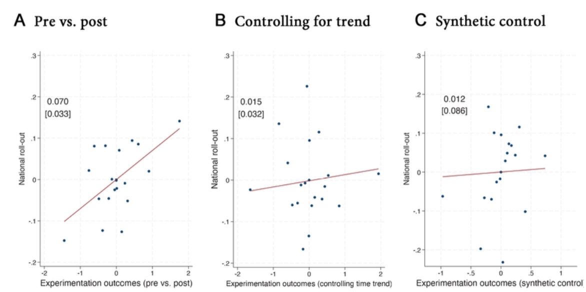

Fig. 4.—These plots visualize the correlations between policy rollout rate and different measures of experimentation effects. In *A*, we construct the simple difference in GDP per capita, among the experiment sites, before and after the policy trial; in *B*, we further control for provincial pretrends in GDP per capita; in *C*, we estimate experimentation effects using a generalized synthetic control approach, where each experimentation site is matched with a weighted average of counties that shares a similar 3-year pretrend, and minister and year fixed effects are included. Coefficients and robust standard errors are reported in the top left corner of each panel.

5.6 Figure 4: what experimentation-effect estimator predicts rollout?

Read:

- Panel A: naive before-after GDP per capita differences among experimentation sites strongly predict whether a policy rolls out nationally.
- Panel B: once one controls for provincial pretrends, the correlation becomes weak.

- Panel C: with generalized synthetic control, the relationship is essentially gone.
- The figure's message is simple and powerful: rollout decisions line up with raw experiment outcomes, not with debiased estimates closer to the true ATE.

5.7 Figure 5: shrinkage from experimentation to national rollout

Read:

- Panel A plots national policy effects against experimentation effects for economic policies. Many points lie below the 45-degree line.
- Panel B plots the distribution of the shrinkage ratio. The highlighted statistic is the headline result: 71.1% of policies have smaller effects after national rollout than during experimentation.
- This figure is the paper's main visual statement about external-validity failure.
- The pattern survives alternative outcome measures such as local fiscal revenue and corrected growth measures.

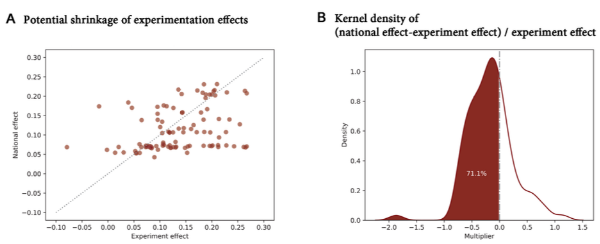

FIG. 5.—These plots demonstrate how policy effects shrink between the experimentation and rollout stages. In *A*, we plot policy effect during national rollout (*y*-axis) against experimentation effect of the same policy (*x*-axis). In *B*, we compute the difference between policy effect during national rollout and policy effect during experimentation, take its ratio over the experiment effect, and plot its distribution.

TABLE 4
SIMILARITY WITH EXPERIMENTATION SITES AND EFFECTS OF POLICY ROLLOUT

	Growth of GDP per Capita		
	(1)	(2)	(3)
	A. Selection of Experimentation Sites		
Mahalanobis distance in socioeconomic conditions	−.004***	−.004***	−.003***
	(.001)	(.001)	(.001)
Number of observations	94,772	94,772	94,772
Number of clusters	2,064	2,064	2,064
Mean of DV	.102	.102	.102
	B. Endogenous Efforts during Experimentation		
Mahalanobis distance in politicians' career incentives	−.001***	−.001***	−.0002
	(.0002)	(.0003)	(.0002)
Number of observations	55,940	55,940	55,940
Number of clusters	1,464	1,464	1,464
Mean of DV	.088	.088	.088
Policy fixed effects	No	No	Yes
Year fixed effects	No	Yes	Yes
County fixed effects	Yes	Yes	Yes

NOTE.—This table investigates how much of a policy's effectiveness at the national rollout stage can be attributed to the site selection and endogenous effort patterns at its experimentation stage. The sample includes all nonexperimentation counties in years that a former policy experiment is being rolled out as a national policy. In panel A, we look at the Mahalanobis distance between experimentation and nonexperimentation counties for a given policy experiment, in terms of their socioeconomic conditions. In panel B, we investigate Mahalanobis distance between the experimentation and nonexperimentation sites in terms of political incentives, where career incentive is measured by the fitted probability of a prefectural party secretary's political promotion, as detailed in app. sec. B.1. The estimated covariance matrix in computing a Mahalanobis distance is fitted by the observed distribution of the data. Mahalanobis distances in both panels are standardized to mean zero and unit variance. Standard errors (shown in parentheses) are clustered at the county level.
*** $p < .01$.

5.8 Table 4: who benefits from national rollout?

Read:

- National policies benefit nonexperiment counties more when those counties look more like the original experimentation sites.
- Similarity in socioeconomic conditions matters strongly.
- Similarity in politicians' career incentives matters too, though somewhat less robustly.

- This means unrepresentative experimentation does not just bias average policy learning; it also creates unequal policy benefits across regions after rollout.

5.9 Appendix evidence worth remembering

Read:

- Complex policies exhibit more positive site selection, while policies with clearer ex ante rollout plans exhibit less positive selection.
- There is little pretrend in fiscal expenditure before experimentation starts.
- The positive-selection problem may have declined slightly over time, but only modestly.
- Social and political unrest both reduces the chance a place is chosen as an experimentation site and lowers the chance that an experimental policy is rolled out nationally.

6 Short summary

- **Conclusion.** China's policy experimentation system is large, structured, and central to national policymaking, but it does not produce clean evidence about average policy effects.
- **Identification insight.** The paper separately identifies three distortions in the experimentation process:
  - nonrepresentative site selection,
  - nonrepresentative politician effort during pilots,
  - naive interpretation of outcomes by the center.
- **Main lesson.** The policies that succeed in experiments are often those tested in richer places, under stronger political incentives, and occasionally under favorable unrelated shocks. When those policies scale nationally, their effects usually shrink.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that China's famous experimentation system is *not* an unbiased policy laboratory.
- The center runs experiments in places that are systematically advantaged, those places exert extra effort because of promotion incentives, and the center does not fully purge unrelated shocks from observed outcomes.
- As a result, experimentation is useful but imperfect: it can facilitate reform, yet it can also generate biased policy learning and distorted national rollout choices.

7.2 Economic intuition

- A policy experiment is never just about the policy. It is also about *where* the policy is tested, *who* implements it, and *how* the evidence is interpreted.
- Richer localities can make policies look better than they really are for the average place.
- Politicians with stronger career incentives can make pilot implementations look better than later national implementations.
- If the center evaluates policies using these inflated pilot outcomes, it will pick too many policies for national rollout and overestimate their likely gains.

7.3 Takeaway

This paper should change how one thinks about Chinese gradualism. The standard view is that experimentation lets the center test ideas locally before scaling them nationally. Wang and Yang show that this mechanism is real but politically contaminated. The contamination is systematic, not incidental: it comes from positive site selection, career-driven effort, and imperfect inference by the center. For research on China, this means that successful pilots should not automatically be read as evidence of strong average treatment effects. For the broader literature on policy learning, the paper's lesson is even more general: experimentation inside a political hierarchy can produce both *better incentives to try policies* and *worse evidence about whether those policies truly scale.*